Clinical Nutrition 39 (2020) 57–66

Contents lists available at ScienceDirect

Clinical Nutrition

journal homepage: http://www.elsevier.com/locate/clnu

Randomized Control Trials

Protein timing has no effect on lean mass, strength and functional capacity gains induced by resistance exercise in postmenopausal women: A randomized clinical trial

Flávia M.S. de Branco [a], Marcelo A.S. Carneiro [b, c], Luana T. Rossato [a], Paula C. Nahas [a], Kely R.C. Teixeira [a], Gersiel N. de Oliveira Jr. [b, c], Fábio L. Orsatti [b, c], Erick P. de Oliveira [a, *]

[a] School of Medicine, Federal University of Uberlandia (UFU), Uberlandia, Minas Gerais, Brazil
[b] Exercise Biology Research Group (BioEx), Federal University of Triangulo Mineiro (UFTM), Uberaba, MG, Brazil
[c] Department of Sport Sciences, Federal University of Triangulo Mineiro, Uberaba, MG, Brazil

ARTICLE INFO

Article history:
Received 18 January 2018
Accepted 9 January 2019

Keywords:
Protein timing
Anabolic window
Muscle mass gain
Older adults
Body composition

SUMMARY

Background & aims: Short-term studies have shown that protein intake immediately post-exercise increases muscle protein synthesis. However, the effect of protein timing (comparing protein intake post-exercise vs. several hours after exercise) on lean mass and strength gains in long-term studies is still not fully elucidated. Thus, the aim of our study was to evaluate the effect of protein timing on lean mass, strength and functional capacity gains induced by resistance exercise in postmenopausal women.
Methods: Thirty-four postmenopausal women (60.9 ± 6.7 years) participated in this double-blind, parallel-group, randomized clinical trial. All individuals performed the same resistance training protocol in the morning, 3 times a week, at 70% of 1-maximum repetition (1-RM), over 8 weeks. Participants were randomly assigned to protein-carbohydrate group (PC) (n = 17), that ingested 30 g of whey protein immediately after exercise and 30 g of maltodextrin in the afternoon; and to carbohydrate-protein group (CP) (n = 17), that ingested 30 g of maltodextrin immediately after exercise and 30 g of whey protein in the afternoon. Lean mass was assessed using dual-energy X-ray absorptiometry, handgrip strength by a dynamometer, and strength was evaluated by 1-RM of bench press and leg extension. One mile walk test was performed to assess the functional capacity.
Results: Both the PC group (37.3 [35.0–39.7] to 38.1 [35.9–40.5] kg) and the CP group (38.2 [36.0–40.5] to 38.8 [36.5–41.3] kg) increased the total lean mass ($p < 0.001$). An increase was also observed in both groups for 1-RM bench press, 1-RM leg extension and handgrip strength ($p < 0.001$). In addition, the time of 1-mile walk test decreased in both groups ($p = 0.019$). No differences were noted for group and time interaction for these variables ($p > 0.05$).
Conclusion: Protein timing has no effect on lean mass, strength and functional capacity gains induced by resistance exercise in postmenopausal women. This trial was registered at ClinicalTrials.gov as NCT03372876.

© 2019 Elsevier Ltd and European Society for Clinical Nutrition and Metabolism. All rights reserved.

1. Introduction

Aging is associated with changes in body composition, such as increased body fat and reduced muscle mass, besides reduction in strength and functional capacity [1,2]. In this context, postmenopausal women, due to estrogen reduction and aging, have an accelerated increase in muscle mass loss [3]. Considering that loss of muscle mass and strength are independently related to mortality in older individuals [4,5], strategies to prevent the muscle mass and strength losses are important. In this way, resistance training and protein intake are non-pharmacological strategies that can reverse or attenuate these effects of aging [6,7]. In relation to protein intake, there are some details that seem to be important for lean mass and strength gains/maintenance, such as the total daily amount [8,9], quality [10,11], distribution during the day [12], dose per meal [13], and timing [14–21].

In respect to protein timing, short-term studies have shown an increase in muscle protein synthesis when protein is ingested after

* Corresponding author. Av. Pará, nº1720 Bloco 2U, Campus Umuarama, 38400-902, Uberlândia, MG, Brazil.
E-mail address: erick_po@yahoo.com.br (E.P. de Oliveira).

https://doi.org/10.1016/j.clnu.2019.01.008
0261-5614/© 2019 Elsevier Ltd and European Society for Clinical Nutrition and Metabolism. All rights reserved.

resistance exercise [22,23]. This phenomenon, known as "anabolic window of opportunity" [24], is a short period that lasts from 45 min to 1 h after the exercise [25]. However, the effect of protein timing on lean mass and strength gains in long term studies is still not elucidated [26]. This effect was evaluated in a meta-analysis performed by Schoenfeld et al. [27], and it was observed that ingesting protein close to exercise promoted a higher lean mass gain when compared to the control group. Nevertheless, the majority of studies included in the meta-analysis did not match total protein intake between groups. The intervention groups ingested more protein during the day (~1.7 g/kg) when compared to control groups (~1.3 g/kg) [27]; therefore, it is not possible to conclude whether the higher lean mass gain occurred due to the timing or to the higher total protein intake. In the same meta-analysis [27], the authors performed a sub-analysis adjusting for total protein intake and the effect of protein timing on lean mass gain was lost, which shows that the higher total protein intake possibly leads to higher lean mass gain, but protein timing does not. However, these results are based on statistical adjustments and randomized controlled clinical trials equalizing the total protein intake during the day are necessary to confirm these data.

In addition, only few studies evaluated the effect of protein timing in older adults [28–33] and only one study was performed in older women [33], whereas no effect was found for lean mass or strength gains [28–33]. There are several ways to evaluate the effect of protein timing, and the only study that evaluated it in older women [33] compared the effect of protein intake pre vs. post-exercise. However, until now, no study evaluated the effect of protein intake post-exercise vs. several hours after exercise in postmenopausal women, which allows to conclude the effect of protein intake immediately after exercise, and also close to exercise on lean mass gain, which is an important knowledge gap to be addressed in this population. Thus, the aim of our study was to evaluate the effect of protein timing (comparing protein intake post-exercise vs. several hours after exercise) on lean mass, strength and functional capacity gains induced by resistance exercise in postmenopausal women. We hypothesized that the timing of protein intake would not lead to additional gains in lean mass, strength and functional capacity when the total protein intake is adequate.

2. Methods

2.1. Research participants

This study was conducted at the Federal University of Uberlandia and at the Federal University of Triangulo Mineiro, Minas Gerais, Brazil. The postmenopausal women were recruited through television advertisements or by direct contact. Inclusion criteria were: postmenopausal women (at least 1 year of cessation of menstruation, self-reported); at least 6 months without performing resistance exercise, and to be disposed to ingest the given supplements twice a day. Those volunteers that were diabetics, had cancer, rheumatoid arthritis, renal and/or liver disease, who were taken anti-inflammatory drugs, who were not available to exercise in the morning and/or were performing hormonal replacement therapy were excluded. Dyslipidemia and hypertension were not considered exclusion criteria. This trial was approved by the Federal University of Uberlandia Research Ethics Committees (protocol number 2.023.127) and registered at ClinicalTrials.gov as NCT03372876. All of the volunteers were previously informed about the experimental procedures and signed a written consent form.

After the disclosure of the study, 123 women were interested in participating, but only 49 were eligible due to the inclusion and exclusion criteria. We performed 1-week familiarization (learning of the exercise movements), and 3 weeks of progressive adaptation of 1–3 sets of 70% of 1-RM (8–12 repetitions) due to the postmenopausal women of the present study being previously untrained. During the adaptation period, 15 participants dropped out of the study, and, after this period, 34 women were randomized (MedCalc® software, version 11.1, Mariakerk, Belgium) into 2 groups: protein-carbohydrate (PC), who ingested 30 g of whey protein immediately after exercise and 30 g of maltodextrin in the afternoon; and carbohydrate-protein group (CP), who ingested 30 g of maltodextrin immediately after exercise and 30 g of whey protein in the afternoon). The individuals were matched randomly by total lean mass values at baseline (after the adaptation period) by an independent and blinded researcher. A flowchart of the progress of this clinical trial is showed in Fig. 1. An a priori sample size for F test (within-between interaction) was calculated using G*Power (v. 3.0.10, Heinrich-Heine-University Düsseldorf, Düsseldorf, Germany). On the basis of a statistical power ($1-\beta$- err prob) of 0.80, an effect size $F = 0.25$, and an overall level of significance of 0.05, a total sample size of 24 individuals were required for this study.

2.2. Study design

This was a randomized double-blind, parallel-group, clinical trial with 8 weeks of follow-up. Dietary habit (three 24-h food recalls) was performed 5 weeks before the beginning of the study. Due to being previously untrained individuals, 1 week of familiarization was performed (−4 week) and, at the end of this week, 1-repetition maximum (1-RM) was evaluated. Then, 3 weeks of adaptation were performed to ensure that the exercise protocol during the supplementation would be done correctly and until concentric failure. Before the beginning of supplementation, body composition, measurements for anthropometry, strength and functional tests were evaluated. After all baseline tests, 8 weeks of protein and carbohydrate supplementation plus resistance exercise were performed. To ensure the correct evaluation of dietary habits, eight 24-h food recalls were performed during the supplementation period. At the end of the study, body composition, measurements for anthropometry, strength and functional tests were replicated (Fig. 2).

2.3. Assessments

2.3.1. Anthropometry

Lohman et al. [34] protocol was used to evaluate body weight and height. Body weight was measured in the morning, with the volunteers wearing lightest clothes, and without shoes. After these measures, Body Mass Index (BMI) was calculated (body weight/height squared).

2.3.2. Body composition

The volunteers arrived at the laboratory after an overnight fast and were instructed to wear light clothing, to be free of metal objects and to ingest 2 L of water in the previous day for standardization of muscle hydration. Total lean mass, leg lean mass, appendicular lean mass, and fat mass were measured using DXA (Lunar iDXA®, GE Healthcare, Madison, WI, USA) and quantified by software Encore version 14.10 (GE Medical Systems, Madison, WI, USA). The same well-trained researcher performed all the exams to avoid measurements error. The Muscle Mass Index (MMI) was calculated by appendicular lean mass (kg) divided by height squared [35].

2.3.3. Strength measurement

To evaluate upper and lower body muscle strength, 1-RM test was performed in bench press (free weight) and leg extension (weight machine) exercises. After the familiarization of the exercise

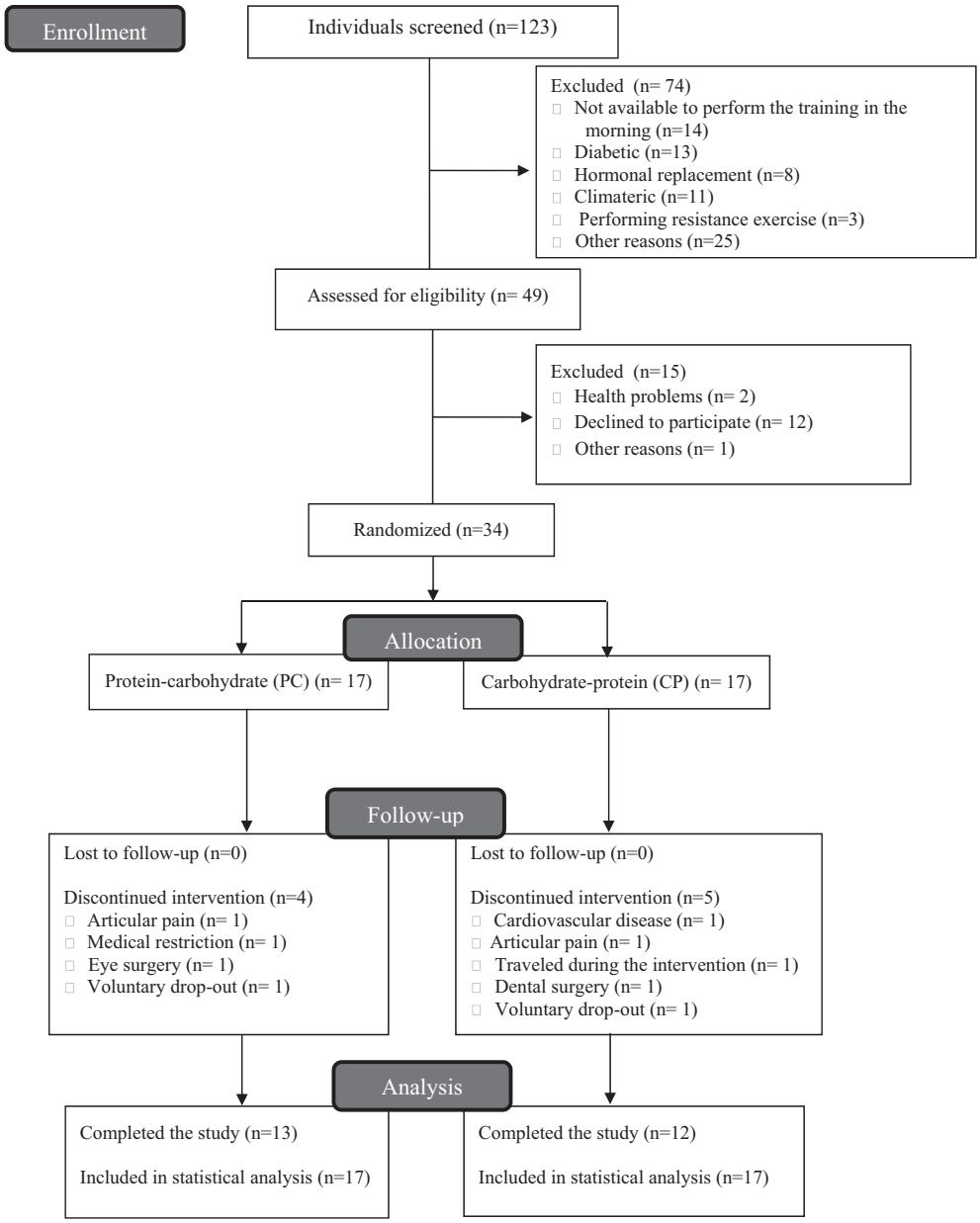

Fig. 1. Flowchart of the progress of the trial.

techniques in 3 sessions, 1-RM was tested and, after 48 h, retested to avoid the underestimation of the load due to the lack of knowledge of the test by the participant. The volunteers performed a walking warm-up for 5 min and then underwent to a specific warm-up that consisted of 15 repetitions with 30% of 1-RM. The load was adjusted, and participants were instructed to perform 10–12 repetitions with 50% of 1-RM after 1 min of rest. After another 1 min of rest, the last warm-up session was performed and consisted of 3–5 repetitions with 80% of 1-RM. When the warm-up was completed, load was increased, and the subjects were encouraged to try to perform 1-RM for the first time. If the participant was not able to make a complete movement or made more than one repetition, 1-RM was tested again after 3 min of rest [36]. A maximum of 3 attempts was made for determination of all 1-RM tests.

Isometric handgrip strength was performed on the right hand using a dynamometer (Jamar®, UK). During the assessment, volunteers were standing with the elbow straight and were instructed to tighten the device after a voice command. The measurement was performed three times with a rest interval of 20 s and the higher value was used [37,38].

2.3.4. Functional capacity measurement

To evaluate the functional capacity, 1-mile walk test was performed on a multi-sport court. The test was performed at baseline (retested after 48 h [39]) and at the end of the study. The result of 1-mile walk test was expressed in minutes.

For sarcopenia diagnosis, the 4-m walk test was performed. The participants should walk as their usual gait speed during daily activities. The walk test was repeated without rest, and the attempt performed in a shorter time was considered to calculate the speed in meters per second [40].

2.3.5. Sarcopenia diagnosis

Sarcopenia diagnosis was applied according to the European Consensus Diagnostic Criteria for Sarcopenia [41]. Muscle Mass Index was considered low when <5.5 kg/m^2; HGS was considered

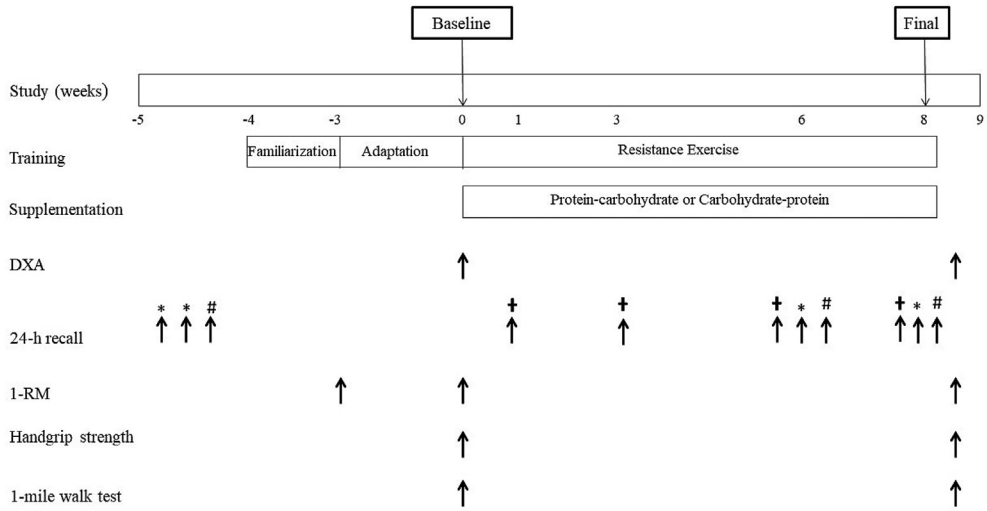

Fig. 2. Schematic overview of study design. * non-training days; #weekend days; † training days.

altered when <20 kg for women; and physical performance, assessed by the 4-m walking test, was considered low when the speed was <0.8 m/s. Participants were classified with sarcopenia when they had reduced muscle mass and decreased strength and/or functional capacity [41].

2.3.6. Dietary assessment

The dietary intake was assessed by 24-h food recalls following the 5-step multiple pass [42]. Three dietary recalls were performed (2 weekdays and 1 weekend day) to evaluate the dietary habits before the familiarization (weeks −5 and −4). To correctly assess dietary habits during the supplementation period, 8 food recalls were performed (weeks 1, 3, 6 and 8) which assessed dietary intake on 4 training days, 2 non-training days, and 2 weekend days (Fig. 2). The food recalls evaluated on weeks −5, 1 and 3 were performed face-to-face and the others via phone call after proper familiarization of size servings and meals names. Calories, carbohydrate, protein, lipid and leucine intake were evaluated. Dietpro® software (version 5.7i, Agromídia Softwares®, Minas Gerais, Brazil) using the United States Department of Agriculture (USDA) [43,44] food composition table was used to perform the analyses of food data.

The distribution of protein and leucine intake throughout the day was evaluated on training days. The meals were divided into: breakfast, morning snack, lunch, afternoon snack, dinner and evening snack according to the time and the meal that was considered by the volunteer.

2.4. Experimental protocol

2.4.1. Resistance training

The resistance exercise was conducted only in the Public Health and Physical Activity Center at the Federal University of Triangulo Mineiro and was performed 3 times a week on non-consecutive days and always in the morning (7 and 8 a.m.) due to the timing supplementation protocol. The exercise protocol followed the recommendations of American College of Sports Medicine [45] for muscle hypertrophy. Before the training session, participants performed 5 min of walking warm-up on a multi-sport court. The resistance exercise was performed for upper and lower limbs, such as bench press (free weight), pec deck (stacked-weight machine), lat pull down (stacked-weight machine), seated low row (stacked-weight machine), leg extension (stacked-weight machine), leg curl (stacked-weight machine), leg press (stacked-weight machine), free weight calf and abdominal. As most women had never performed resistance exercise, before onset the supplement intervention, 3 weeks of progressive adaptation of 1–3 sets of 70% of 1-RM (8–12 repetitions) was performed. The resistance exercise was performed with 3 sets of 8–12 repetitions (70% 1-RM). As volunteers increased their strength, the loads were readjusted weekly to maintain the training zone of 8–12 repetitions of 70% of 1-RM until the concentric failure. The minimum acceptable training adherence was 70% [46].

2.4.2. Supplementation

The PC group ingested 30 g of whey protein (Hilmar™ 9010 Instantized Whey Protein Isolate) immediately after exercise (8–9 a.m.) and 30 g of carbohydrate (Maltodextrin Cargill®) at ~3 p.m. The CP group ingested 30 g of carbohydrate immediately after exercise (8–9 a.m.) and 30 g of whey protein at ~3 p.m. The supplements were ingested every day over 8 weeks at the same time of the day (8–9 a.m. and 3 p.m.). The participants received the supplements in an opaque packing from a blinded researcher after each training session and the supplements were taken in an opaque bottle. Furthermore, the participants received once a week the supplements that should be ingested in the afternoon period for training days; and in the morning and in the afternoon for non-training days. All individuals were instructed to return the empty package to determine the adherence to supplementation. All the supplements were distributed in individual servings containing exactly 30 g of whey or maltodextrin. Both supplements (whey and maltodextrin) were isocaloric (only 4.8 kcal of difference) and strawberry flavored. The information about the supplements composition is showed in Supplemental Table 1.

On training days, the subjects were instructed to refrain from consuming protein food sources during the morning, but they could ingest carbohydrate and fat food sources. The lunch was eaten no less than 2 h (individuals ingested ~3 h) after the cessation of each exercise session (Fig. 3).

2.5. Statistics

All analyzes were performed using SPSS software (version 20.0, IBM Corp, New York, NY, USA). Baseline characteristics and the distribution of protein and leucine during the day were analyzed using Generalized Estimating Equations (GEE) using unstructured

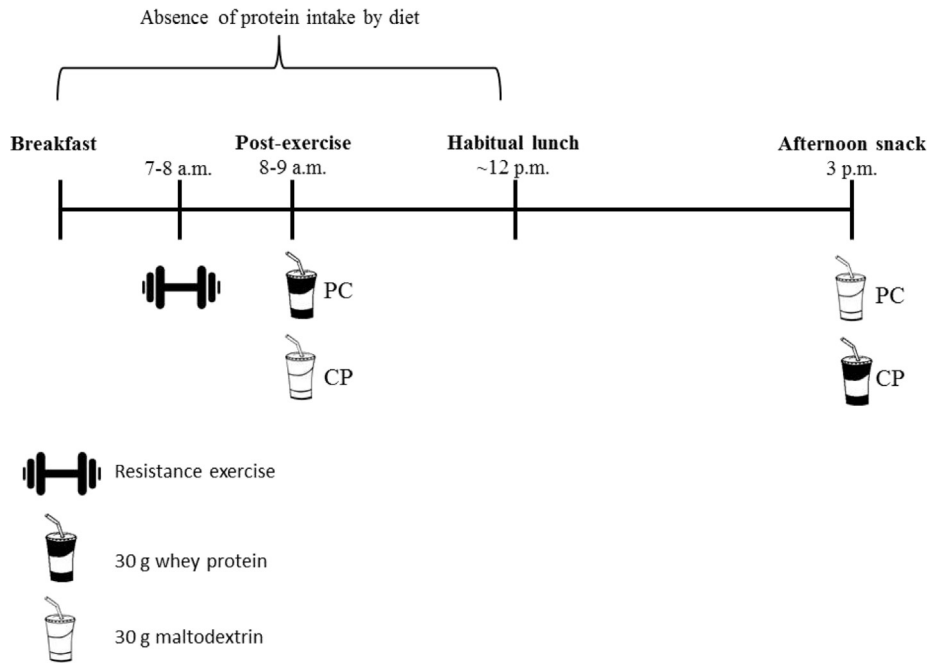

Fig. 3. Supplementation protocol on training days. PC: protein-carbohydrate group. CP = carbohydrate-protein group.

correlation and maximum likehood estimation. Moreover, changes in body composition, strength and dietary intake were analyzed by an intention-to-treat analysis using GEE and Sequencial Sidak post-hoc adjusted for the adherence of protein supplementation. Significance was accepted when *P-value* was <0.05. All values are described as means and Wald 95% confidence intervals.

3. Results

3.1. Baseline characteristics

At baseline, no differences were observed between groups for age, body weight, BMI, lean mass (total, appendicular, leg and trunk), fat mass, strength (bench press, leg extension and handgrip), 1-mile walk test; calories, protein, lipids and carbohydrate intake (Table 1). In addition, none of the volunteers was diagnosed with sarcopenia.

3.2. Supplement and training adherence

The adherence to supplementation (PC = 82.2 [70.4–96.1] and CP = 72.9 [58.2–88.8] %; p = 0.315) and to training (PC = 83.2 [71.4–96.7] ad CP 72.8 [59.5–89.2] %; p = 0.306) were similar between the groups, evaluating all the individuals that entered in the study (n = 34) and considering the intention-to-threat analysis. When only the participants that completed all the protocol (n = 25) were evaluated, there were also no differences between the groups in relation to the supplementation adherence (PC = 95.6 [94.2–97.1] and CP = 91.6 [87.5–96.0] %; p = 0.084) and to training adherence (PC = 95.2 [91.8–98.7] and CP 91.7 [86.1–97.7] %; p = 0.315). The adherence to the supplementation on training days was 100% for all individuals that completed the study (n = 25).

3.3. Dietary intake during intervention

No differences were found in dietary intake from food sources over the time in both groups. When protein and carbohydrate supplementation were included in dietary intake, an increase in calories, carbohydrate in grams; protein in g/kg, g and %; and leucine (g) was observed; whereas lipids (g) and carbohydrate (%) did not change, and lipids (%) decreased in both groups (Table 2).

3.4. Distribution of protein and leucine on training days

For all meals, no difference was observed in habitual dietary intake between the groups on training days. The protein intake from food sources before exercise at breakfast was 2.8 [2.1–3.9] and 3.4 [2.5–4.7] g; and at morning snack (post-training) was 2.2 [1.5–3.2] and 1.6 [1.0–2.5] g for PC and CP (p > 0.05), respectively. When the supplement was included after exercise, an increase in protein intake was observed in PC group compared to CP group at morning snack (29.3 [28.6–30.1] vs. 1.6 [1.0–2.5] g, p < 0.001; respectively). At lunch, both groups showed similar protein intake (PC = 26.3 [20.3–34.0] vs. CP = 29.1 [24.2–35.0] g; p = 0.531). At afternoon snack, the protein intake exclusively from food was similar between the groups, but CP increased the protein intake due to protein supplementation (CP = 33.4 [31.8–35.0] vs. PC = 3.7 [2.7–5.3] g; p < 0.001). The protein intake at dinner and evening snack was similar between the groups (Table 3).

For leucine, a small and insignificant amount was ingested from food sources before the exercise at breakfast and after exercise at morning snack. When the supplement was taken after exercise, an increase in leucine taken at morning snack was observed in PC group. At lunch, both groups showed a similar leucine intake. At afternoon snack, both groups ingested similar amounts of leucine from food sources and after the addition of the supplement, CP group had a greater intake of leucine. The leucine intake at dinner and evening snack was similar between the groups (Table 3).

3.5. Muscle strength and functional capacity

Both groups increased 1-RM in bench press and in leg extension exercises over the time (p < 0.001). Handgrip strength increased in both groups over the study (p < 0.001) and a reduction in time (minutes) to finish the 1-mile walk test was observed at the end of the study in both groups (p = 0.019). No difference was observed in group × time interaction for all the variables (Table 4).

Table 1
Baseline characteristics of the participants.

Variables	PC ($n = 17$)	CP ($n = 17$)	*p-value*
Age, *y*	60.5 [57.7–63.3]	61.4 [58.0–65.0]	0.680
Anthropometrics			
Body weight, *kg*	69.1 [62.2–76.5]	71.3 [65.7–77.3]	0.646
Height, *m*	1.58 [1.56–1.62]	1.59 [1.57–1.62]	0.774
BMI, kg/m^2	27.9 [25.2–30.8]	27.6 [25.7–29.7]	0.878
Body Composition			
Total lean mass, *kg*	37.3 [35.0–39.7]	38.2 [36.0–40.5]	0.597
Leg lean mass, *kg*	12.9 [11.8–14.2]	13.5 [12.5–14.6]	0.519
Trunk lean mass, *kg*	17.2 [16.3–18.2]	17.5 [16.6–18.4]	0.703
MMI, kg/m^2	6.9 [6.4–7.5]	6.8 [6.4–7.3]	0.750
Total fat mass, *kg*	29.4 [24.5–35.4]	30.6 [27.2–34.5]	0.727
Total fat mass, *%*	42.5 [38.2–47.2]	43.8 [41.8–46.0]	0.578
Trunk fat mass, *kg*	15.1 [12.3–18.4]	15.3 [13.4–17.4]	0.919
Strength			
Bench press 1-RM (kg)	29.2 [27.1–31.3]	29.3 [26.2–32.7]	0.952
Leg extension 1-RM (kg)	30.0 [26.5–33.8]	30.7 [26.5–35.7]	0.783
Handgrip strength (right hand), *kg*	27.0 [25.6–28.5]	27.4 [24.8–30.2]	0.819
Functional capacity			
1-mile walk test, *minutes*	15.7 [14.7–16.8]	15.3 [14.7–15.9]	0.520
Dietary intake			
Calories, *kcal*	1589 [1371–1842]	1690 [1499–1906]	0.523
Carbohydrate, *g*	196 [163–236]	200 [176–229]	0.844
Carbohydrate, *%*	49.5 [45.4–54.0]	48.0 [44.5–51.8]	0.595
Protein, *g*	62.4 [52.9–73.5]	69.1 [58.7–81.3]	0.386
Protein, *g/kg*	0.9 [0.7–1.1]	1.0 [0.8–1.2]	0.355
Protein, *%*	15.6 [14.2–17.2]	16.5 [14.8–18.3]	0.454
Lipids, *g*	58.4 [49.9–68.4]	60.9 [52.0–71.4]	0.717
Lipids, *%*	33.5 [31.1–36.0]	32.7 [29.5–36.2]	0.718

Notes: PC, protein-carbohydrate group; CP, carbohydrate-protein group. BMI: body weight index; MMI: muscle mass index, was calculated by appendicular lean mass (kg) divided by height squared. All values are means and 95% confidence-intervals. Significantly different between groups $p < 0.05$.

3.6. Anthropometric and body composition

No changes were observed in body weight ($p = 0.069$), but the BMI increased ($p = 0.018$) over the time in both groups. Both PC group and CP group increased the total lean mass after the intervention ($p < 0.001$), but no differences between the groups ($p = 0.643$) and group × time interaction ($p = 0.408$) were found. Both groups increased the leg lean mass ($p < 0.001$) with no differences between groups ($p = 0.510$) and no group × time interaction ($p = 0.930$). An increase in muscle mass index was also observed in both groups ($p < 0.001$), with no differences between groups ($p = 0.839$) and no time × group interaction ($p = 0.826$). Total fat mass (kg) did not change in PC and CP groups over the time (Table 4).

4. Discussion

The main finding of the present study was that protein timing had no effect on lean mass, strength and functional capacity gains induced by 8 weeks of resistance exercise in postmenopausal women who ingested an adequate amount of protein. Both PC and CP groups performed the resistance exercise in the morning and all the individuals were not allowed to ingest protein before exercise. In addition, while the PC group ingested ~30 g of protein immediately after exercise, the CP group only ingested protein after ~3 h after exercise (at lunch; from food sources) and the protein supplementation was taken in the afternoon (at 3:00 p.m.). To the best of our knowledge, this is the first study showing these results in previously untrained postmenopausal women.

Table 2
Intake of calories, macronutrients, and leucine before and during the study.

Variables	PC ($n = 17$)		CP ($n = 17$)		*p-value*		
	Pre	During	Pre	During	Group	Time	Time x Group
Energy intake, *kcal*	1589 [1371–1842]	1446 [1228–1702]	1690 [1499–1906]	1662 [1478–1869]	0.253	0.239	0.412
Energy intake including supplement, *kcal*	1589 [1371–1842]	1686 [1467–1938]	1690 [1499–1906]	1902 [1717–2107]	0.264	0.044	0.506
Carbohydrate intake, *g*	196 [163–236]	184 [150–226]	200 [175–229]	219 [196–244]	0.338	0.819	0.190
Carbohydrate intake including supplement, *g*	196 [163–236]	214 [180–255]	200 [176–229]	249 [226–274]	0.361	0.005	0.234
Carbohydrate intake, *%*	49.5 [45.4–54.0]	50.2 [45.9–54.8]	48.0 [44.5–51.8]	53.4 [49.1–57.9]	0.759	0.057	0.140
Carbohydrate intake including supplement, *%*	49.5 [45.4–54.0]	50.2 [46.7–54.0]	48.0 [44.5–51.8]	52.9 [49.3–56.8]	0.826	0.055	0.157
Protein intake, *g*	62.4 [52.9–73.5]	57.6 [46.8–70.9]	69.1 [58.7–81.3]	64.8 [56.6–74.2]	0.285	0.287	0.905
Protein intake including supplement, *g*	62.4 [52.9–73.5]	85.4 [74.3–98.2]	69.1 [58.7–81.3]	92.7 [84.3–102]	0.286	<0.001	0.859
Protein intake, *g/kg*	0.9 [0.7–1.1]	0.9 [0.6–1.2]	1.0 [0.8–1.2]	0.9 [0.8–1.0]	0.671	0.312	0.673
Protein intake including supplement, *g/kg*	0.9 [0.7–1.1]	1.3 [1.0–1.6]	1.0 [0.8–1.2]	1.3 [1.2–1.4]	0.691	<0.001	0.474
Protein intake, *%*	15.6 [14.2–17.2]	15.8 [14.4–17.4]	16.5 [14.8–18.3]	15.6 [14.5–16.7]	0.798	0.608	0.492
Protein intake including supplement, *%*	15.6 [14.2–17.2]	20.5 [19.2–21.8]	16.5 [14.8–18.3]	19.6 [18.6–20.7]	0.999	<0.001	0.267
Leucine intake, *g*	4.5 [3.7–5.4]	4.2 [3.4–5.3]	4.7 [3.9–5.6]	4.5 [3.9–5.3]	0.591	0.457	0.836
Leucine intake including supplement, *g*	4.5 [3.7–5.4]	7.2 [6.3–8.1]	4.7 [3.9–5.6]	7.5 [6.8–8.2]	0.608	<0.001	0.983
Lipids, *g*	58.4 [49.9–68.4]	52.0 [44.7–59.3]	60.9 [52.0–71.4]	56.5 [46.9–68.0]	0.520	0.103	0.679
Lipids intake, *%*	33.5 [31.1–36.0]	32.6 [30.5–34.8]	32.7 [29.5–36.2]	29.9 [27.6–32.4]	0.200	0.146	0.429
Lipids intake including supplement, *%*	33.5 [31.1–36.0]	27.6 [25.9–29.3]	32.7 [29.5–36.2]	26.0 [23.8–28.5]	0.345	<0.001	0.687

Notes: PC, protein-carbohydrate group; CP, carbohydrate-protein group. All values are means and 95% confidence-intervals. Significantly different between groups $p < 0.05$.

Table 3
Distribution of protein and leucine on training days (supplementation timing).

Meal	Groups		*p-value*
	PC ($n = 17$)	CP ($n = 17$)	
Protein			
Breakfast	2.8 [2.1–3.9]	3.4 [2.5–4.7]	0.445
Morning snack	2.2 [1.5–3.2]	1.6 [1.0–2.5]	0.277
Morning snack including supplement	29.3 [28.6–30.1]	1.6 [1.0–2.5]	<0.001
Lunch	26.3 [20.3–34.0]	29.1 [24.2–35.0]	0.531
Afternoon snack	3.7 [2.7–5.3]	5.3 [4.3–6.7]	0.095
Afternoon snack including supplement	3.7 [2.7–5.3]	33.4 [31.8–35.0]	<0.001
Dinner	18.3 [14.7–22.9]	16.9 [13.4–21.2]	0.610
Evening snack	2.6 [1.5–4.6]	2.5 [1.3–4.5]	0.885
Leucine			
Breakfast	0.18 [0.13–0.36]	0.22 [0.16–0.31]	0.383
Morning snack	0.13 [0.09–0.20]	0.11 [0.07–0.17]	0.552
Morning snack including supplement	3.0 [2.9–3.1]	0.11 [0.07–0.17]	<0.001
Lunch	2.0 [1.5–2.6]	2.1 [1.7–2.5]	0.914
Afternoon snack	0.25 [0.17–0.38]	0.38 [0.28–0.50]	0.114
Afternoon snack including supplement	0.25 [0.17–0.38]	3.3 [3.1–3.4]	<0.001
Dinner	1.36 [1.09–1.69]	1.20 [0.95–1.51]	0.443
Evening snack	0.20 [0.10–0.38]	0.23 [0.12–0.44]	0.738

All values are means and 95% confidence-intervals. Significantly different between groups $p < 0.05$.

The effect of protein timing on lean mass gain is not totally elucidated [26] and it has been studied in young adults [14–21,47–51] and older adults [28–33]. These studies evaluated the effect of protein timing in several ways, such as comparing the protein intake pre vs. post-exercise [14,18,19,31,33,47,49], pre and post vs. several hours after exercise [17,48], only post-exercise [29,51]; and post-exercise vs. several hours after exercise [11,50]. Of all these studies, only one [33] evaluated the effect of protein timing exclusively in postmenopausal women. In this study, the authors compared the effect of protein intake pre vs. post-resistance exercise in older women and no difference was observed for lean mass gain [33]. The absence of difference between the protein intake pre vs. post-exercise can probably be explained due to the duration of protein synthesis after intake of an adequate amount of protein. It is well known that protein synthesis lasts from 2 to 5 h after protein intake [25], and considering that a resistance exercise protocol generally lasts from 45 to 60 min per session, when the protein is ingested before or after exercise, both groups may present higher protein synthesis induced by protein intake after resistance exercise. However, this type of protein timing is different from that performed in the present study, since we compared the effect of protein intake immediately after exercise vs. ~3 h after exercise, whereas both groups did not intake protein ~8 h before exercise (overnight fast). Therefore, we have two main findings: firstly, the protein intake after exercise did not promote higher lean mass gain, such as other studies [33,47]; secondly, we showed that the protein intake close to resistance exercise did not promote additional effects on lean mass gain. These results can possibly be explained because resistance exercise induces an increase in protein synthesis that lasts 24–48 h [52,53], and not only few hours after exercise. Therefore, every meal containing protein during the day can contribute for lean mass gain, which shows that the total protein intake seems to be more important than the protein timing, that was also confirmed by a meta-analysis performed by Schoenfeld et al. [27].

Few studies that evaluated the effect of protein timing equalized the total protein intake [17,28,33,47,48]. However, only in one study [33] the resistance exercise induced lean mass gain, whereas in other studies no changes were observed on lean mass [47,48], and even in one study a lean mass loss [28] was noted. Therefore, with exception of the Nabuco et al. (2018) study [33], the other studies [17,28,47,48] are not totally comparable with our findings because it would be unlikely that the protein timing would induce to additional lean mass gain since the resistance training was not effective in promoting muscle hypertrophy. It is known that resistance exercise is the main intervention to stimulate lean mass and strength gains [46,54], and protein supplementation without resistance exercise effects does not seem to increase lean mass [55].

The postmenopausal women of the present study ingested 30 g of isolated whey protein (27.6 g of protein) and 30 g of maltodextrin.

Table 4
Body composition, strength and functional tests values according to moments and groups.

Variables	PC		CP		*p-value*		
	Pre (n = 17)	Post (n = 17)	Pre (n = 17)	Post (n = 17)	Group	Time	Time x group
Body weight (kg)	69.3 [62.6–76.6]	69.6 [62.7–77.3]	70.9 [65.6–76.7]	71.8 [66.2–77.9]	0.678	0.069	0.478
Body Mass Index (kg/m^2)	27.9 [25.2–30.8]	28.1 [25.4–31.1]	27.6 [25.7–29.7]	27.8 [25.8–29.9]	0.851	0.018	0.504
Total lean mass (kg)	37.3 [35.0–39.7]	38.1 [35.9–40.5]	38.2 [36.0–40.5]	38.8 [36.5–41.3]	0.643	<0.001	0.408
Leg lean mass (kg)	12.9 [11.8–14.2]	13.3 [12.2–14.5]	13.5 [12.5–14.6]	13.8 [12.8–14.9]	0.510	<0.001	0.930
Trunk lean mass (kg)	17.2 [16.3–18.2]	17.5 [16.6–18.5]	17.5 [16.6–18.4]	17.5 [16.5–18.6]	0.845	0.036	0.212
Total fat mass (kg)	29.4 [24.5–35.4]	29.3 [24.4–35.3]	30.6 [27.2–34.5]	30.5 [27.0–34.3]	0.727	0.434	0.981
Trunk fat mass (kg)	15.1 [12.3–18.4]	15.2 [12.5–18.5]	15.3 [13.4–17.4]	15.3 [13.3–17.5]	0.946	0.538	0.613
Muscle mass index (kg/m^2)	6.9 [6.4–7.5]	7.1 [6.6–7.6]	6.8 [6.4–7.3]	7.0 [6.6–7.5]	0.839	<0.001	0.826
Bench press RM (kg)	29.2 [27.1–31.3]	33.8 [31.4–36.6]	29.3 [26.2–32.7]	34.8 [31.3–38.8]	0.800	<0.001	0.539
Leg extension RM (kg)	30.0 [26.5–33.8]	33.8 [31.0–37.0]	30.7 [26.5–35.7]	33.3 [29.4–37.8]	0.947	<0.001	0.318
Handgrip strength (kg)	27.0 [25.6–28.5]	28.3 [26.5–30.2]	27.4 [24.8–30.2]	29.3 [26.6–32.1]	0.664	<0.001	0.479
1-mile walk test (minutes)	15.7 [14.7–16.8]	15.6 [14.7–16.5]	15.3 [14.7–15.9]	15.0 [14.3–15.7]	0.333	0.019	0.796

Notes: PC, protein-carbohydrate group; CP, carbohydrate-protein group. All values are means and 95% confidence intervals. Significantly different between groups $P < 0.05$.

We chose whey as protein source since this is a protein rich in leucine, that promotes higher muscle protein synthesis than other protein sources [56]. We did not add carbohydrate in the protein beverage because when adequate doses of protein are ingested there are no further augment in protein synthesis with addition of carbohydrate [57]. In addition, this protein dose was chosen because we evaluated postmenopausal women, and it is recommended to consume ~30 g of high quality protein per meal [12] or ~0.4 g/kg/meal [13] for maximum muscle protein synthesis stimulation in older adults. In the present study, it was offered 27.6 g or ~0.43 g/kg/meal of protein (27.6 g of protein divided by the mean weight of the individuals); therefore, we offered the protein dose that stimulate the maximum muscle protein synthesis, which allowed us to evaluate the protein timing using the ideal dose of protein for this population. Only one study offered a similar dose than the present study [33], whereas other studies that evaluated the protein timing in older adults offered lower doses, such as 10 g [58], 20 g [31], and 21 g [28] of protein. In addition, in the present study, both supplements were blinded with the same flavor (strawberry flavored), and there were no differences between the groups regarding the correct answers and errors about which supplements were being ingested (data not shown), which confirms the correct blinding in the present study. Thus, our supplementation protocol was adequate to evaluate the effect of protein timing.

The volunteers were instructed to abstain the protein intake from food sources in the morning on training days during the intervention. This recommendation was important to exclude the effects of protein intake from food sources on the supplementation timing effects. It is important to remember that this recommendation did not seem to influence the habitual protein intake of these individuals, since the low protein intake in the morning is already a habit of Brazilian postmenopausal women [46] and was also observed in the present study at baseline (data not shown). Despite this orientation, a consumption of small amount of protein in both groups were observed at breakfast (~3 g) and at morning snack (~1.5 g) due to protein contained in breads and fruits, that were frequently ingested in these two meals. However, these amounts of protein seem to be insignificant for the muscle protein synthesis stimulus and it was not a confounding factor to evaluate the protein timing effects. The dietary intake in the present study was assessed by eleven 24 h food recalls, being 3 at baseline and 8 during the study (4 on training days, 2 on the weekends, and 2 on non-training days). Therefore, our study provided a reliable evaluation of dietary intake since most studies only evaluate the intake before and after the intervention [15,17,30,31,48,58], and as whey protein intake can increase satiety [59], it would be possible that the protein supplementation may have decreased the protein and calorie intake by diet during these interventions. However, in our study, both PC and CP groups maintained the habitual intake of calories and protein during the intervention by diet; and when the supplements were added in dietary intake, an increase in calories, protein, carbohydrate, and leucine intake in both groups was found, with no differences between the interventions. In addition, besides both groups ingested the same amount of total protein (~1.3 g/kg/day), the distribution of protein during the day (by diet) was the same between groups during the intervention, which is an important dietary control to evaluate the effect of the protein timing.

We performed 1-week familiarization (learning of the exercise movements) and 3-weeks of adaptation of the resistance training due to the postmenopausal women of the present study being previously untrained; and also because they should perform each set until muscle concentric failure over the study. These 3-weeks of adaptation were important to promote motor learning related to neural adaptations, because the increase in muscular strength occurs faster than the lean mass gain [60]. In addition, it was recently shown that during the first 3 weeks of resistance training a high muscle damage is induced and the protein synthesis is only correlated with hypertrophy after the attenuations of the muscle damage [61]. Therefore, we only initiated the protein supplementation after this adaptation. Both groups increased lean mass (~0.7 kg), strength, and functional capacity similarly after the intervention, which showed that 8-weeks of resistance training were efficient to promote benefits. The resistance exercise-induced gains of these parameters are in accordance with other studies that evaluated postmenopausal women [3,46,62,63]. Thus, considering that fat-free mass declines 2–3% per decade from 30 to 70 years of age [64], our resistance exercise protocol reverted approximately 5 years of fat-free mass loss, which is a great impact on health of the participants. The functional capacity was assessed by 1-mile walk test. A longer test was chosen, instead of 4-m walk test, since short tests are more commonly used in frail older adults [65]. As we evaluated postmenopausal women without sarcopenia, the 1-mile test may present a reliable evaluation of functional capacity in these individuals. No effect of protein timing was observed on functional capacity, but the resistance exercise was efficient to improve the walking test, as was also observed by other studies [62,66]. The volunteers performed test and retest, whereas the retest values were used for functional capacity and strength evaluation. The test-retest was performed to avoid improvements in these tests due to learning effects.

No changes in body fat were observed in both groups, which shows that the individuals did not perform a caloric restriction, that could attenuate the lean mass gain [67]. In addition, even with an increase in caloric intake due to protein and carbohydrate supplementation, there was no fat mass gain, that also seems to be important for optimal lean mass gain in postmenopausal women [36]. An increase in BMI was noted at the end of the study in both groups; however, it did not occur due to increase in body fat, but in response to lean mass gain.

The present study has limitations. We performed a short follow-up time (8 weeks) and future studies evaluating the effect of protein timing in longer durations are needed. Another limitation is the absence of a control group without protein supplementation. In addition, the physical activity during the free-living conditions was not evaluated. As strengths, the detailed control of the habitual diet allows us to conclude the correct effect of the protein timing on lean mass and strength gains. We controlled the total protein intake, as well as the protein distribution and dose per meal in both groups. In addition, we provided the supplementation on both training days and non-training days.

In conclusion, our results show that protein timing has no effect on lean mass, strength and functional capacity gains induced by resistance exercise in postmenopausal women. Therefore, it does not seem to be important to ingest protein immediately after exercise to promote greater benefits than those already induced by the resistance exercise, at least in postmenopausal women.

Funding

No funding was received for the preparation of this article.

Authorship

FMSB participated in collection and interpretation of the data, performed statistical analysis, and wrote the manuscript; MASC, LTR, PCN and GNO participated in collection, analysis, and interpretation of the data. KRCT performed statistical analysis, collaborated in the collection and interpretation of the data; FLO carried out the conception and design of the study and revised the manuscript. EPO carried out the conception and design of the study, participated in the interpretation of the data, wrote, and

contributed with the revision of the manuscript. All authors read and approved the final manuscript.

Conflict of interest

All the authors declare no conflicts of interest.

Acknowledgments

We are thankful to “Fundação de Amparo à Pesquisa de MG” (FAPEMIG) for the financial support.

Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.clnu.2019.01.008.

References

[1] Fielding RA, Vellas B, Evans WJ, Bhasin S, Morley JE, Newman AB, et al. Sarcopenia: an undiagnosed condition in older adults. Current consensus definition: prevalence, etiology, and consequences. International working group on sarcopenia. J Am Med Dir Assoc 2011;12:249–56.
[2] Kim TN, Choi KM. Sarcopenia: definition, epidemiology, and pathophysiology. J Bone Metabol 2013;20:1–10.
3] Orsatti FL, de Oliveira EP, Burini RC. Relationship between plasma hormones and anthropometric measures of muscle mass in postmenopausal women. Handb Anthrop 2012:1481–90.
[4] Rantanen T, Volpato S, Ferrucci L, Heikkinen E, Fried LP, Guralnik JM. Handgrip strength and cause-specific and total mortality in older disabled women: exploring the mechanism. J Am Geriatr Soc 2003;51:636–41.
[5] Wannamethee SG, Shaper AG, Lennon L, Whincup PH. Decreased muscle mass and increased central adiposity are independently related to mortality in older men. Am J Clin Nutr 2007;86:1339–46.
[6] Cermak NM, Res PT, de Groot LC, Saris WH, van Loon LJ. Protein supplementation augments the adaptive response of skeletal muscle to resistance-type exercise training: a meta-analysis. Am J Clin Nutr 2012;96:1454–64.
[7] Lixandrao ME, Damas F, Chacon-Mikahil MP, Cavaglieri CR, Ugrinowitsch C, Bottaro M, et al. Time course of resistance training-induced muscle hypertrophy in the elderly. J Strength Condit Res 2016;30:159–63.
[8] Phillips SM, Chevalier S, Leidy HJ. Protein "requirements" beyond the rda: implications for optimizing health. Appl Physiol Nutr Metab 2016;41:565–72.
[9] Tieland M, Dirks ML, van der Zwaluw N, Verdijk LB, van de Rest O, de Groot LC, et al. Protein supplementation increases muscle mass gain during prolonged resistance-type exercise training in frail elderly people: a randomized, double-blind, placebo-controlled trial. J Am Med Dir Assoc 2012;13:713–9.
[10] Phillips SM. The impact of protein quality on the promotion of resistance exercise-induced changes in muscle mass. Nutr Metab (Lond) 2016;13:64.
[11] Hartman JW, Tang JE, Wilkinson SB, Tarnopolsky MA, Lawrence RL, Fullerton AV, et al. Consumption of fat-free fluid milk after resistance exercise promotes greater lean mass accretion than does consumption of soy or carbohydrate in young, novice, male weightlifters. Am J Clin Nutr 2007;86:373–81.
[12] Paddon-Jones D, Rasmussen BB. Dietary protein recommendations and the prevention of sarcopenia. Curr Opin Clin Nutr Metab Care 2009;12:86–90.
[13] Moore DR, Churchward-Venne TA, Witard O, Breen L, Burd NA, Tipton KD, et al. Protein ingestion to stimulate myofibrillar protein synthesis requires greater relative protein intakes in healthy older versus younger men. J Gerontol A Biol Sci Med Sci 2015;70:57–62.
[14] Andersen LL, Tufekovic G, Zebis MK, Crameri RM, Verlaan G, Kjaer M, et al. The effect of resistance training combined with timed ingestion of protein on muscle fiber size and muscle strength. Metab Clin Exp 2005;54:151–6.
[15] Bird SP, Tarpenning KM, Marino FE. Independent and combined effects of liquid carbohydrate/essential amino acid ingestion on hormonal and muscular adaptations following resistance training in untrained men. Eur J Appl Physiol 2006;97:225–38.
[16] Candow DG, Burke NC, Smith-Palmer T, Burke DG. Effect of whey and soy protein supplementation combined with resistance training in young adults. Int J Sport Nutr Exerc Metabol 2006;16:233–44.
[17] Cribb PJ, Hayes A. Effects of supplement timing and resistance exercise on skeletal muscle hypertrophy. Med Sci Sports Exerc 2006;38:1918–25.
[18] Willoughby DS, Stout JR, Wilborn CD. Effects of resistance training and protein plus amino acid supplementation on muscle anabolism, mass, and strength. Amino acids 2007;32:467–77.
[19] Hulmi JJ, Kovanen V, Selanne H, Kraemer WJ, Hakkinen K, Mero AA. Acute and long-term effects of resistance exercise with or without protein ingestion on muscle hypertrophy and gene expression. Amino acids 2009;37:297–308.
[20] Josse AR, Tang JE, Tarnopolsky MA, Phillips SM. Body composition and strength changes in women with milk and resistance exercise. Med Sci Sports Exerc 2010;42:1122–30.
[21] Walker TB, Smith J, Herrera M, Lebegue B, Pinchak A, Fischer J. The influence of 8 weeks of whey-protein and leucine supplementation on physical and cognitive performance. Int J Sport Nutr Exerc Metabol 2010;20:409–17.
[22] Biolo G, Tipton KD, Klein S, Wolfe RR. An abundant supply of amino acids enhances the metabolic effect of exercise on muscle protein. Am J Physiol 1997;273:E122–9.
[23] Tipton KD, Ferrando AA, Phillips SM, Doyle Jr D, Wolfe RR. Postexercise net protein synthesis in human muscle from orally administered amino acids. Am J Physiol 1999;276:E628–34.
[24] van Loon LJ. Is there a need for protein ingestion during exercise? Sports Med 2014;44(Suppl. 1):S105–11.
[25] Moore DR, Tang JE, Burd NA, Rerecich T, Tarnopolsky MA, Phillips SM. Differential stimulation of myofibrillar and sarcoplasmic protein synthesis with protein ingestion at rest and after resistance exercise. J Physiol 2009;587:897–904.
[26] Naderi A, de Oliveira EP, Ziegenfuss TN, Willems MET. Timing, optimal dose and intake duration of dietary supplements with evidence-based use in sports nutrition. J Exerc Nutrition Biochem 2016;20:1–12.
[27] Schoenfeld BJ, Aragon AA, Krieger JW. The effect of protein timing on muscle strength and hypertrophy: a meta-analysis. Sports Nutr Rev J 2013;10:53.
[28] Wycherley TP, Noakes M, Clifton PM, Cleanthous X, Keogh JB, Brinkworth GD. Timing of protein ingestion relative to resistance exercise training does not influence body composition, energy expenditure, glycaemic control or cardiometabolic risk factors in a hypocaloric, high protein diet in patients with type 2 diabetes. Diabetes Obes Metab 2010;12:1097–105.
[29] Godard MP, Williamson DL, Trappe SW. Oral amino-acid provision does not affect muscle strength or size gains in older men. Med Sci Sports Exerc 2002;34:1126–31.
[30] Candow DG, Chilibeck PD, Facci M, Abeysekara S, Zello GA. Protein supplementation before and after resistance training in older men. Eur J Appl Physiol 2006;97:548–56.
[31] Verdijk LB, Jonkers RA, Gleeson BG, Beelen M, Meijer K, Savelberg HH, et al. Protein supplementation before and after exercise does not further augment skeletal muscle hypertrophy after resistance training in elderly men. Am J Clin Nutr 2009;89:608–16.
[32] Eliot KA, Knehans AW, Bemben DA, Witten MS, Carter J, Bemben MG. The effects of creatine and whey protein supplementation on body composition in men aged 48 to 72 years during resistance training. J Nutr Health Aging 2008;12:208–12.
[33] Nabuco H, Tomeleri C, Sugihara Junior P, Fernandes R, Cavalcante E, Antunes M, et al. Effects of whey protein supplementation pre- or post-resistance training on muscle mass, muscular strength, and functional capacity in pre-conditioned older women: a randomized clinical trial. Nutrients 2018;10:563.
[34] Lohman TG, Roche AF, Martorell R. Anthropometric standardization reference manual. 1988.
[35] Baumgartner RN, Koehler KM, Gallagher D, Romero L, Heymsfield SB, Ross RR, et al. Epidemiology of sarcopenia among the elderly in New Mexico. Am J Epidemiol 1998;147:755–63.
[36] Orsatti FL, Nahas EA, Orsatti CL, de Oliveira EP, Nahas-Neto J, da Mota GR, et al. Muscle mass gain after resistance training is inversely correlated with trunk adiposity gain in postmenopausal women. J Strength Condit Res 2012;26:2130–9.
[37] Haidar SG, Kumar D, Bassi RS, Deshmukh SC. Average versus maximum grip strength: which is more consistent? J Hand Surg Br Eur Vol 2004;29:82–4.
[38] Reis BCA, de Branco FMS, Pessoa DF, Barbosa CD, dos Reis AS, de Medeiros LA, et al. Phase angle is positively associated with handgrip strength in hospitalized individuals. Top Clin Nutr 2018;33:127–33.
[39] Francis P, Mc Cormack W, Lyons M, Jakeman P. Age-group differences in the performance of selected tests of physical function and association with lower extremity strength. J Geriatr Phys Ther 2019;42:1–8.
[40] Guralnik JM, Ferrucci L, Simonsick EM, Salive ME, Wallace RB. Lower-extremity function in persons over the age of 70 years as a predictor of subsequent disability. N Engl J Med 1995;332:556–61.
[41] Cruz-Jentoft AJ, Baeyens JP, Bauer JM, Boirie Y, Cederholm T, Landi F, et al. Sarcopenia: European consensus on definition and diagnosis: report of the european working group on sarcopenia in older people. Age Ageing 2010;39:412–23.
[42] Conway JM, Ingwersen LA, Vinyard BT, Moshfegh AJ. Effectiveness of the us department of agriculture 5-step multiple-pass method in assessing food intake in obese and nonobese women. Am J Clin Nutr 2003;77:1171–8.
[43] USDA. Food composition databases. United States Department of Agriculture; 2006 [slightly revised May].
[44] USDA. Food composition databases. United States Department of Agriculture; 2007.
[45] ACSM. American college of sports medicine's health-related physical fitness assessment manual. 2013.
[46] Rossato LT, Nahas PC, de Branco FMS, Martins FM, Souza AP, Carneiro MAS, et al. Higher protein intake does not improve lean mass gain when compared with rda recommendation in postmenopausal women following resistance exercise protocol: a randomized clinical trial. Nutrients 2017;9.

[47] Schoenfeld BJ, Aragon A, Wilborn C, Urbina SL, Hayward SE, Krieger J. Pre-versus post-exercise protein intake has similar effects on muscular adaptations. Peer J 2017;5:e2825.
[48] Hoffman JR, Ratamess NA, Tranchina CP, Rashti SL, Kang J, Faigenbaum AD. Effect of protein-supplement timing on strength, power, and body-composition changes in resistance-trained men. Int J Sport Nutr Exerc Metabol 2009;19:172–85.
[49] Antonio J, Sanders MS, Ehler LA, Uelmen J, Raether JB, Stout JR. Effects of exercise training and amino-acid supplementation on body composition and physical performance in untrained women. Nutrition 2000;16:1043–6.
[50] Hoffman JR, Ratamess NA, Kang J, Falvo MJ, Faigenbaum AD. Effects of protein supplementation on muscular performance and resting hormonal changes in college football players. J Sports Sci Med 2007;6:85–92.
[51] Rankin JW, Goldman LP, Puglisi MJ, Nickols-Richardson SM, Earthman CP, Gwazdauskas FC. Effect of post-exercise supplement consumption on adaptations to resistance training. J Am Coll Nutr 2004;23:322–30.
[52] Tang JE, Perco JG, Moore DR, Wilkinson SB, Phillips SM. Resistance training alters the response of fed state mixed muscle protein synthesis in young men. Am J Physiol Regul Integr Comp Physiol 2008;294:R172–8.
[53] Phillips SM, Tipton KD, Aarsland A, Wolf SE, Wolfe RR. Mixed muscle protein synthesis and breakdown after resistance exercise in humans. Am J Physiol 1997;273:E99–107.
[54] Morton RW, Murphy KT, McKellar SR, Schoenfeld BJ, Henselmans M, Helms E, et al. A systematic review, meta-analysis and meta-regression of the effect of protein supplementation on resistance training-induced gains in muscle mass and strength in healthy adults. Br J Sports Med 2018;52:376–84.
[55] Tieland M, Franssen R, Dullemeijer C, van Dronkelaar C, Kyung Kim H, Ispoglou T, et al. The impact of dietary protein or amino acid supplementation on muscle mass and strength in elderly people: individual participant data and meta-analysis of rct's. J Nutr Health Aging 2017;21:994–1001.
[56] Tang JE, Phillips SM. Maximizing muscle protein anabolism: the role of protein quality. Curr Opin Clin Nutr Metab Care 2009;12:66–71.
[57] Koopman R, Beelen M, Stellingwerff T, Pennings B, Saris WH, Kies AK, et al. Coingestion of carbohydrate with protein does not further augment post-exercise muscle protein synthesis. Am J Physiol Endocrinol Metab 2007;293:E833–42.
[58] Esmarck B, Andersen JL, Olsen S, Richter EA, Mizuno M, Kjaer M. Timing of postexercise protein intake is important for muscle hypertrophy with resistance training in elderly humans. J Physiol 2001;535:301–11.
[59] Sousa GT, Lira FS, Rosa JC, de Oliveira EP, Oyama LM, Santos RV, et al. Dietary whey protein lessens several risk factors for metabolic diseases: a review. Lipids Health Dis 2012;11:67.
[60] Carroll TJ, Riek S, Carson RG. Neural adaptations to resistance training: implications for movement control. Sports Med 2001;31:829–40.
[61] Damas F, Phillips SM, Libardi CA, Vechin FC, Lixandrão ME, Jannig PR, et al. Resistance training-induced changes in integrated myofibrillar protein synthesis are related to hypertrophy only after attenuation of muscle damage. J Physiol 2016;594:5209–22.
[62] Nunes PRP, Oliveira AA, Martins FM, Souza AP, Orsatti FL. Effect of resistance training volume on walking speed performance in postmenopausal women: a randomized controlled trial. Exp Gerontol 2017;97:80–8.
[63] Orsatti FL, Maestá N, de Oliveira EP, Nahas Neto J, Burini RC, Nunes PRP, et al. Adding soy protein to milk enhances the effect of resistance training on muscle strength in postmenopausal women. J Diet Suppl 2018;15:140–52.
[64] Chodzko-Zajko WJ, Proctor DN, Fiatarone Singh MA, Minson CT, Nigg CR, Salem GJ, et al. American college of sports medicine position stand. Exercise and physical activity for older adults. Med Sci Sports Exerc 2009;41:1510–30.
[65] Volpato S, Cavalieri M, Sioulis F, Guerra G, Maraldi C, Zuliani G, et al. Predictive value of the short physical performance battery following hospitalization in older patients. J Gerontol Series A Biol Sci Med Sci 2011;66:89–96.
[66] Barbosa CD, Costa JG, Giolo JS, Rossato LT, Nahas PC, Mariano IM, et al. Isoflavone supplementation plus combined aerobic and resistance exercise do not change phase angle values in postmenopausal women: a randomized placebo-controlled clinical trial. Exp Gerontol 2018. https://doi.org/10.1016/j.exger.2018.08.007 (in press).
[67] Weinheimer EM, Sands LP, Campbell WW. A systematic review of the separate and combined effects of energy restriction and exercise on fat-free mass in middle-aged and older adults: implications for sarcopenic obesity. Nutr Rev 2010;68:375–88.